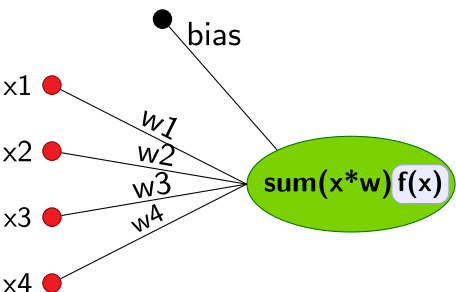


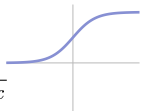
Structure of a single neuron:



Activation Functions

Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



tanh

$$\tanh(x)$$



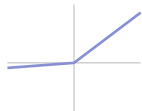
ReLU

$$\max(0, x)$$



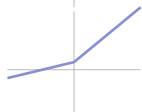
Leaky ReLU

$$\max(0.1x, x)$$



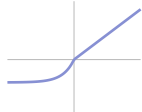
Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$



ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



A 3-layers fully connected neural network (DNN)

